Part A: Order of Operations

Evaluate each expression completely.

1. $(-9+(-2))^2 \div (-11)$
2. $(5+3)^2 \div (-2)$
3. $(-5)^2 - (-20) \cdot 6$
4. $(-4+12)^2 \div 2$
5. $(9+3)^2 \div (-8)$
6. $(2+14)^2 \div (-8)$
7. $18 - 8 \cdot ((-4)+18)$
8. $8^2 - 4 \cdot 2$
9. $(-2+15)^2 \div 1$
10. $24 - 4 \cdot ((-9)+24)$
11. $31 - 8 \cdot ((-5)+31)$
12. $14 - 3 \cdot ((-2)+14)$
13. $8^2 - 3 \cdot 4$
14. $40 - 4 \cdot (5+40)$
15. $(-7)^3 - 17 \cdot 9$

Part B: Negative Number Operations

Evaluate each expression completely.

1. $-12 \cdot (-4)$
2. $-9 \cdot 11$
3. $8 \div (-4)$
4. $-64 \div 8$
5. $-20 \div 10$
6. $-60 \div 12$

7. $-34 + (-9) \cdot 15$

8. $37 + 8 \cdot 6$

9. $11 + (-11) \cdot 6$

10. $196 \div 14 - (-32)$

11. $28 + (-15) \cdot (-14)$

12. $-27 + 9 \cdot (-10)$

13. $25 \div 5 - (-5)$

14. $72 \div (-6) - (-20)$

15. $33 \div 3 - 40$

Part C: Solving Linear Equations

Solve each equation for x.

1. $9x - 10 = -\frac{13}{4}$

2. $12x + 12 = -54$

3. $4x + 4 = -4$

4. $9x - 12 = 6$

5. $5x - 8 = -3$

6. $8(8x - 11) + 3 = -213$

7. $5(4x + 9) + 4 = -191$

8. $8(x - 4) + 5 = 5$

9. $3(2x + 3) - 10 = -25$

10. $13x - 4 = -10x + 19$

11. $-6x + 14 = 11x - \frac{23}{2}$

12. $-8x + 13 = 15x + 151$

13. $-10x - 10 = -15x + 45$

14. $\dfrac{7x+56}{4} = \dfrac{14}{8}$

15. $\dfrac{2x-20}{12} = \dfrac{2x}{4}$

Part D: Linear Inequalities

Solve each inequality for x. Remember to reverse the inequality sign when you multiply or divide by a negative number.

1. $5x \leq 35$
2. $x + 9 < 1$
3. $x - 8 < -8$
4. $3x - 10 > -13$
5. $8x - 8 \leq 80$
6. $-2x + 1 \geq 21$
7. $2x + 9 > -9$
8. $-12x + 7 < -125$
9. $-12x - 11 \leq -35$
10. $4x - 5 > 13x + 76$
11. $-3x - 11 < x + 49$
12. $10x + 12 \geq -15x + 62$
13. $\frac{5}{2}x - 11 > -2x - 47$
14. $-3x + 3 > \frac{5}{3}x + 59$
15. $2x - 6 > \frac{9}{2}x - 31$

Part E: Slope-Intercept Form

Rewrite each equation in slope-intercept form if needed, then identify the slope m and the y-intercept b.

1. $y = \frac{9}{2}x + 4$
2. $y = 3x - 5$
3. $y = 7x + 1$
4. $y = 4x - 9$
5. $y = 7x - 2$
6. $y = 9x - 5$
7. $y = 9x + 5$
8. $4y = 32x + 16$
9. $20x - 4y = 32$
10. $20x - 4y = 24$
11. $16x - 4y = -32$
12. $3y = 18x - 6$
13. $-9x + 3y = 24$
14. $4y = 14x - 24$
15. $-24x + 3y = -15$

Part F: Exponent Rules

Simplify each expression. Write your answer using only positive exponents.

1. $3x^3 \cdot 2x^6$
2. $2x^9 \cdot 5x^2$
3. $5x^5 \cdot 6x^2$

4. $\dfrac{2x^7}{x^3}$

5. $\dfrac{2x^{11}}{x^7}$

6. $\dfrac{4x^{11}}{x^2}$

7. $(6x^2)^2$

8. $(8x^4)^2$

9. $(2x^6)^2$

10. $6 \cdot (10^3)^{-3}$

11. $5 \cdot 3^{-1}$

12. $8 \cdot 6^{-4}$

13. $(3x^4y^2)^2 \cdot 2x^9y^1$

14. $(5x^{-3})^3 \cdot x^{13}$

15. $\left(\dfrac{4x^4}{2x^2}\right)^2$

Part G: Square and Cube Roots

Simplify each root completely. If the radicand is not a perfect square or cube, use prime factorization to pull out every factor you can.

1. $\sqrt{9}$

2. $\sqrt{100}$

3. $\sqrt{64}$

4. $\sqrt[3]{27}$

5. $\sqrt[3]{8}$

6. $\sqrt{124}$

7. $\sqrt{92}$

8. $\sqrt{148}$

9. $\sqrt{8}$

10. $\sqrt{40}$

11. $\sqrt{75}$

12. $\sqrt{18}$

13. $\sqrt[3]{112}$

14. $\sqrt[3]{40}$

15. $\sqrt[3]{136}$

Answer Key

Part A: Order of Operations

1. -11
2. -32
3. 145
4. 32
5. -18
6. -32
7. -94
8. 56
9. 169
10. -36
11. -177
12. -22
13. 52
14. -140
15. -496

Part B: Negative Number Operations

1. 48
2. -99
3. -2
4. -8
5. -2
6. -5
7. -169
8. 85
9. -55
10. 46
11. 238
12. -117
13. 10
14. 8
15. -29

Part C: Solving Linear Equations

1. $x = \frac{3}{4}$
2. $x = -\frac{11}{2}$
3. $x = -2$
4. $x = 2$
5. $x = 1$
6. $x = -2$
7. $x = -12$
8. $x = 4$
9. $x = -4$
10. $x = 1$
11. $x = \frac{3}{2}$
12. $x = -6$
13. $x = 11$
14. $x = -7$
15. $x = -5$

Part D: Linear Inequalities

1. $x \leq 7$
2. $x < -8$
3. $x < 0$
4. $x > -1$
5. $x \leq 11$
6. $x \leq -10$
7. $x > -9$
8. $x > 11$
9. $x \geq 2$
10. $x < -9$
11. $x > -15$
12. $x \geq 2$
13. $x > -8$
14. $x < -12$
15. $x < 10$

Part E: Slope-Intercept Form

1. $m = \frac{9}{2}, \quad b = 4$
2. $m = 3, \quad b = -5$
3. $m = 7, \quad b = 1$
4. $m = 4, \quad b = -9$
5. $m = 7, \quad b = -2$
6. $m = 9, \quad b = -5$
7. $m = 9, \quad b = 5$
8. $m = 8, \quad b = 4$
9. $m = 5, \quad b = -8$
10. $m = 5, \quad b = -6$
11. $m = 4, \quad b = 8$
12. $m = 6, \quad b = -2$
13. $m = 3, \quad b = 8$
14. $m = \frac{7}{2}, \quad b = -6$
15. $m = 8, \quad b = -5$

Part F: Exponent Rules

1. $6x^9$
2. $10x^{11}$
3. $30x^7$
4. $2x^4$
5. $2x^4$
6. $4x^9$
7. $36x^4$
8. $64x^8$
9. $4x^{12}$
10. $\frac{3}{500000000}$
11. $\frac{5}{3}$
12. $\frac{1}{162}$
13. $18x^{17}y^5$
14. $125x^4$
15. $4x^4$

Part G: Square and Cube Roots

1. 3
2. 10
3. 8
4. 3
5. 2
6. $2\sqrt{31}$
7. $2\sqrt{23}$
8. $2\sqrt{37}$
9. $2\sqrt{2}$
10. $2\sqrt{10}$
11. $5\sqrt{3}$
12. $3\sqrt{2}$
13. $2\sqrt[3]{14}$
14. $2\sqrt[3]{5}$
15. $2\sqrt[3]{17}$